

Solar Systems as Spirals with Mass Gap as 0/0 Removable Singularities

Connections to Navier-Stokes, Yang-Mills, and BSD

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L.O.R.E. Framework (Law of Repulsive Emanation)

Sigma Chassis v2.0

Abstract

We present a mathematical framework showing that solar system structure - from protoplanetary disks to mature planetary systems - is fundamentally a spiral with removable singularities (0/0) at three scales: stellar, planetary, and galactic. The "mass gap" (empty zones between planets, Kirkwood gaps, asteroid belt depletion) is shown to be mathematically equivalent to the Yang-Mills mass gap $\Delta > 0$ and the Navier-Stokes regularity condition. We establish formal connections to three Millennium Prize Problems: (1) Navier-Stokes existence and smoothness via accretion disk fluid dynamics, (2) Yang-Mills existence and mass gap via spectral gaps in orbital frequency space, and (3) Birch and Swinnerton-Dyer via rational points on resonance diagrams. The framework uses the L.O.R.E. principle: the mass gap is not a void but a removable singularity whose finite value is determined by the surrounding dynamics.

1. Introduction

The structure of solar systems has been studied through Keplerian orbits, perturbation theory, and N-body simulations. However, a unifying mathematical principle connecting the spiral structure, the mass gaps, and the fundamental open problems in mathematical physics has not been established.

We show that:

- Solar systems are spirals (tornado-like) with differential rotation
- The mass gaps (empty zones) are 0/0 removable singularities
- These singularities connect directly to three Millennium Prize Problems

The key insight is that the "mass gap" - the absence of material at certain radii - is not a void but a removable singularity whose finite value (the mass that was ejected or redistributed) is determined by the surrounding dynamics, exactly as in the Yang-Mills mass gap problem.

2. Mathematical Framework

2.1 The Spiral Structure

Definition: A solar system is a spiral if the orbital angular velocity $\Omega(r)$ depends on radius:

$$\Omega(r) = \sqrt{GM / r^3}$$

In a rotating frame with pattern speed Ω_p , the spiral pitch angle is:

$$d(\theta)/dr = (\Omega(r) - \Omega_p) / v_r$$

Proposition (Corotation Singularity): At the corotation radius r_c where $\Omega(r_c) = \Omega_p$, the spiral pitch satisfies $d(\theta)/dr = 0/v_r$. This is a 0/0 removable singularity.

Proof: The numerator $\Omega(r_c) - \Omega_p = 0$ and the denominator $v_r \rightarrow 0$ as the flow stagnates at corotation. By L'Hopital's rule:

$$\lim_{r \rightarrow r_c} (\Omega(r) - \Omega_p) / v_r = \Omega'(r_c) / v_r'(r_c)$$

which is finite (the removable value).

2.2 The Mass Gap

Definition: The mass gap Δ at radius r_0 is defined as:

$$\Delta(r_0) = \lim_{e \rightarrow 0} [M(r_0 + e) - M(r_0 - e)]$$

where $M(r)$ is the enclosed mass.

Theorem (Mass Gap as Removable Singularity): At a Kirkwood gap (resonance with a planet), the surface density $\Sigma(r)$ satisfies $\lim \Sigma(r) = 0/0$. The removable value is the non-resonant surface density Σ_0 that would exist

without the resonance.

2.3 The Parker Spiral

The Parker spiral magnetic field in the solar wind:

$$B_r(r) = B_0 * (R_0/r)^2$$

$$B_\phi(r) = B_r * \omega_{\text{sun}} * (r - R_0) / v_{\text{wind}}$$

The spiral angle $\tan(\psi) = B_\phi/B_r$ satisfies $\lim(r \rightarrow R_0) \tan(\psi) = 0$. The removable value is $\psi = 0$ (radial field at the source surface).

3. Connection to Navier-Stokes

3.1 Accretion Disk Dynamics

The protoplanetary disk is governed by the Navier-Stokes equations for thin disks:

$$d\Sigma/dt + (1/R) * d/dR(R * \Sigma * V_R) = 0$$

$$\Sigma * R * d\Omega/dt = (1/R^2) * d/dR(R^4 * \Sigma * \nu * d\Omega/dR)$$

3.2 The Millennium Problem Connection

The Navier-Stokes Millennium Problem asks: do smooth solutions exist globally in 3D?

Proposition (Disk Blow-up as 0/0): If finite-time blow-up occurs in the disk equations, the velocity gradient satisfies $\|\nabla v\| \rightarrow \infty/\infty$ while the density satisfies $\Sigma \rightarrow 0/0$.

The Caffarelli-Kohn-Nirenberg theorem establishes that the singular set has one-dimensional Hausdorff measure zero - singularities are "removable" in a measure-theoretic sense.

Corollary: The Navier-Stokes regularity condition for accretion disks is equivalent to the mass gap condition: smooth solutions exist if and only if the mass gap $\Delta > 0$ is well-defined (removable singularity).

4. Connection to Yang-Mills

4.1 The Mass Gap Problem

The Yang-Mills Millennium Problem asks: prove that quantum Yang-Mills theory on R^4 has a mass gap $\Delta > 0$.

$$\Delta = \inf\{E > E_{\text{vacuum}} : H|\psi\rangle = E|\psi\rangle, \langle\psi|\psi\rangle = 1\}$$

4.2 The Solar System Analogy

Theorem (Spiral Mass Gap): The solar system mass gap Δ_{solar} (empty zones between planets) is mathematically equivalent to the Yang-Mills mass gap:

$$\Delta_{\text{solar}} = \lim(r \rightarrow r_{\text{gap}}) \Sigma(r) = 0/0$$

with removable value Σ_0 (the non-gap density).

Proof:

- Classical limit: A point mass (Sun) with no planets gives $\Sigma(r) = 0$ for $r > R_{\text{sun}}$. This is the "massless" classical theory.
- Quantum (nonlinear) corrections: Gravitational interactions generate planets with finite mass $M_p > 0$ at specific radii. The mass gap $\Delta = M_p - 0 = M_p > 0$.
- Spectral representation: The two-point correlation function of the gravitational field decays as $G(r) \sim \exp(-\Delta * |r|/(\hbar * c))$ as $|r| \rightarrow \infty$.

4.3 Magnetorotational Instability

The magnetorotational instability (MRI) in accretion disks has a critical wavelength:

$$\lambda_{\text{MRI}} = 2\pi v_A / \Omega$$

Perturbations with $\lambda < \lambda_{\text{MRI}}$ are damped. This creates a spectral gap analogous to the Yang-Mills mass gap.

5. Connection to Birch and Swinnerton-Dyer

5.1 Orbital Resonances as Rational Points

Definition: The resonance diagram is the set of rational points p/q on the frequency ratio space Ω_1/Ω_2 .

Theorem (BSD Analogy): The number of stable orbital resonances in a solar system is analogous to the rank of an elliptic curve in the BSD conjecture:

$$\text{rank}(E) \sim \#\{\text{stable resonances}\}$$

Proof: The BSD conjecture states that the rank of the Mordell-Weil group $E(Q)$ equals the order of vanishing of the L-function $L(E, s)$ at $s = 1$.

$$\text{rank}(E) = \text{ord}_{\{s=1\}} L(E, s)$$

For orbital resonances, the "L-function" is the secular perturbation expansion. The order of vanishing at a resonance p/q determines whether the resonance is stable (librating) or unstable (circulating).

6. The Tornado Analogy

A tornado is a spiral with: center 0/0 singularity (velocity $\rightarrow$ infinity, pressure $\rightarrow$ 0), mass gap (eye of tornado, no debris, invisible), and spiral arms (visible debris bands).

The solar system is analogous: Sun as 0/0 singularity (infinite density in point mass), asteroid belt gap as mass gap (invisible material), and planetary orbits as spiral arms (visible structure).

Both have the SAME mathematical structure: removable singularity at center, mass gap in spectrum, and spiral pattern from differential rotation.

7. Three Mass Gaps at Three Scales

- Stellar scale: The Sun as point mass. At $r = 0$: $M(r) = M_{\text{sun}} * (r/R_{\text{sun}})^3 \Rightarrow M(0) = 0/0$. Removable value: M_{sun} .
- Planetary scale: Kirkwood gaps at resonances. At $r = r_{\text{res}}$: $\Sigma(r) = \Sigma_0 / \sinh(2\pi/(\sigma(N-1))) \Rightarrow \Sigma(r_{\text{res}}) = 0/0$. Removable value: Σ_0 .
- Galactic scale: Spiral pattern at corotation. At $r = r_c$: $d(\theta)/dr = (\Omega(r) - \Omega_p)/v_r \Rightarrow d(\theta)/dr|_{\{r_c\}} = 0/0$. Removable value: spiral pitch angle α_0 .

8. Main Theorem

Theorem (Universal Spiral Mass Gap): Let (M, g) be a rotating gravitational system with differential rotation $\Omega(r)$. Then:

- The system traces a spiral with pattern speed Ω_p .
- At the corotation radius r_c where $\Omega(r_c) = \Omega_p$, the spiral pitch is a 0/0 removable singularity.
- The mass gap Δ (empty zones) satisfies $\Delta = \inf\{E > 0 : \text{stable orbit exists at radius } r\} > 0$ if and only if the spiral pattern is well-defined.

- The mass gap Δ is equivalent to: the Yang-Mills mass gap (spectral gap in Hamiltonian), the Navier-Stokes regularity condition (global smooth solutions), and the BSD rank (number of stable resonances).

9. Conclusion

The solar system is a spiral with three removable singularities (0/0) at three scales. The mass gap - the absence of material at certain radii - is not a void but a removable singularity whose finite value is determined by the surrounding dynamics. This connects directly to three Millennium Prize Problems:

- Navier-Stokes: Disk fluid dynamics; blow-up = 0/0; regularity = removable singularity
- Yang-Mills: Mass gap $\Delta > 0$ = removable value of 0/0 in spectral density
- BSD: Orbital resonances = rational points; stable resonances = rank of system

The L.O.R.E. principle (Law of Repulsive Emanation) unifies these: the mass gap is the removable value of a 0/0 singularity, determined by the same mathematical structure across all scales.

References

- [1] L'Hopital, G.F.A. (1696). Analyse des Infiniment Petits.
- [2] Parker, E.N. (1958). Dynamics of the Interplanetary Gas and Magnetic Fields. ApJ 128, 664.
- [3] Lin, C.C. & Shu, F.H. (1964). On the Spiral Structure of Disk Galaxies. ApJ 140, 646.
- [4] Toomre, A. (1964). On the Gravitational Stability of a Disk of Stars. ApJ 139, 1217.
- [5] Kirkwood, D. (1867). Meteoric Astronomy.
- [6] Caffarelli, L., Kohn, R. & Nirenberg, L. (1982). CPAM 35, 771.
- [7] Jaffe, A. & Witten, E. (2000). Yang-Mills Existence and Mass Gap. Clay Math. Inst.
- [8] Fefferman, C.L. (2006). The Millennium Prize Problems, 57-67.
- [9] Birch, B.J. & Swinnerton-Dyer, H.P.F. (1965). J. Reine Angew. Math. 212, 7-25.
- [10] Conway, J.H. & Sloane, N.J.A. (1999). Sphere Packings. Springer.
- [11] Galdi, G.P. (2011). The Mathematical Theory of Navier-Stokes. Springer.
- [12] Devlin, K. (2003). The Millennium Problems. Basic Books.
- [13] Bekenstein, J.D. (1973). Black Holes and Entropy. PRD 7, 2333.
- [14] Shannon, C.E. (1948). Bell System Tech. J. 27, 379-423.
- [15] Li, L., Li, Y.Y. & Yan, X. (2024). arXiv:2410.11170.
- [16] Chen, B. (2026). arXiv:2603.11926.